

Zhiyu Wu

Zhiyuwu2@illinois.edu • <https://xzzwzy.github.io/> • 7345968166 • 410 N Lincoln Ave

EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

M.S. Computer Science – Thesis Based

University of Michigan

Ann Arbor, MI

B.S.E. Computer Engineering (Dual Degree)

August 2022 – May 2024

Shanghai Jiao Tong University

Shanghai, China

B.S.E. Electrical and Computer Engineering (Dual Degree) GPA: 3.75 / 4.00

Sept. 2020 – August 2024

Research Interest: LLMs, ML system infrastructure, systems for LLMs

Coursework: Computer Architecture, Operating Systems, Computer Network, Machine Learning, Compiler

WORK EXPERIENCE

Research Internship, Bytedance Data-AML-Machine Learning System

San Jose, CA

Supervisor: Xiao Yu

May 2025 - present

- Developed comprehensive Jupyter notebook and web frontend toolkit integrating the complete AI model training workflow from dataset profiling, parallelism optimization and simulation, SFT/RL deployment to production inference.
- Built performance simulator for distributed LLM training with focus on memory modeling, accurately predicting GPU memory consumption patterns and inter-node communication overhead.

RESEARCH EXPERIENCE

Cross-layer KV Cache Parallelism for LLM Serving at Scale

Champaign, IL

Supervisor: Fan Lai

August 2024 - present

- Identified KV cache as a growing memory and latency bottleneck in large-scale LLM serving systems.
- Developed novel cross-layer parallelism framework combining token-wise, layer-wise, and multi-GPU parallelism for hybrid KV cache restoration.
- Implemented SLO-aware scheduling to manage I/O contention across requests, batches, and models while maintaining performance guarantees.
- Addressed limitations of existing systems in long-context scenarios and concurrent restoration workloads.
- Designed cross-layer management system that optimally balances computation and communication for KV cache restoration at cluster scale.

Application-aware LLM Serving with Mixed SLO Requirements

Champaign, IL

Supervisor: Fan Lai

July 2024 – May 2025

- Identified diverse SLO requirements across three LLM request patterns: latency-sensitive, throughput-intensive, and collective requests with varying performance needs.
- Developed SLO-aware scheduler that maximizes service gain by allocating just enough bandwidth to meet individual SLOs while preserving residual capacity for other requests.
- Implemented hybrid approach using quantile regression forests for response length estimation and dependency-graph matching for collective request coordination.
- Created service density-based prioritization algorithm that balances requests across different SLO constraints without starving low-priority workloads.
- Achieved 1.3×-8.3× improvement in service gain and 4.0X-10.3X improvement in SLO goodput compared to existing systems across diverse workloads and models.

Defining and Enhancing Quality-of-Experience in LLM-Based Text Streaming Services

Ann Arbor, MI

Supervisor: Mosharaf Chowdhury

May 2023 – April 2024

- Identified that in LLM text-streaming services, systems must generate faster than user reading speed to enhance user experience, addressing gaps in previous metrics.
- Defined Quality of Experience (QoE) in LLM serving by tracking each step of text generation and monitoring the overall user experience throughout the entire streaming process.
- Formulated the problem as a knapsack optimization and developed a scheduling algorithm to maximize QoE in online LLM serving.

- Built Andes, an LLM serving system on top of vLLM, integrating the scheduling algorithm to enhance QoE in real-time LLM services.
- Co-authored the paper “Andes: Defining and Enhancing Quality-of-Experience in LLM-Based Text Streaming Services” as the second author.

PUBLICATIONS

- [Tempo: Application-aware LLM Serving with Mixed SLO Requirements](#); Preprint, 2025; Wei Zhang*, **Zhiyu Wu***, Yi Mu, Banruo Liu, Myungjin Lee, Fan Lai
- [Andes: Defining and Enhancing Quality-of-Experience in LLM-Based Text Streaming Services](#); Preprint, 2024; Jiachen Liu, **Zhiyu Wu**, Jae-Won Chung, Fan Lai, Myungjin Lee, Mosharaf Chowdhury
- [The ML. ENERGY Benchmark: Toward Automated Inference Energy Measurement and Optimization](#); Preprint, 2025; Jae-Won Chung, Jiachen Liu, Jeff J Ma, Ruofan Wu, Oh Jun Kweon, Yuxuan Xia, **Zhiyu Wu**, Mosharaf Chowdhury

PROJECT EXPERIENCE

Symbiotic Lab/ML.ENERGY.LEADERBOARD Team

Ann Arbor, MI

Developer

May 2023 – Sept. 2023

- Developed the ML.ENERGY Leaderboard, an open-source platform for benchmarking the energy efficiency and NLP performance of LLM models.
- Defined performance metrics and implemented scripts for optimized batched inference to ensure accurate measurement.
- Contributed to the online Chatbot Arena which gathers data on models' energy consumption and performance.

Toy Operating System

Ann Arbor, MI

- Created a toy operating system with physical memory and disk management.
- Implemented read-write locks using mutexes to manage multi-threading.
- Developed virtual memory management with a page table and a network file server using sockets.
- Built a custom file system for networked access.

Out-of-order Execution Pipeline for the MIPS R10K Microprocessor

Shanghai, China

- Developed an out-of-order execution pipeline with six stages on the MIPS R10K microprocessor.
- Implemented key components including register renaming, reservation station, reorder buffer, and a common data bus for enhanced parallelism.
- Applied Tomasulo's algorithm for dynamic scheduling and reducing pipeline stalls.
- Added a Load Store Queue and a Branch Target Buffer to further optimize execution efficiency and improve instruction throughput.

Video Streaming via CDN

Ann Arbor, MI

- Developed a proxy server for handling video streaming across multiple clients and servers, ensuring scalability and reliability.
- Implemented adaptive bitrate streaming to minimize buffering and enhance user experience based on real-time network conditions.
- Used DNS load balancing with round-robin and distance-based server selection, utilizing Dijkstra's algorithm to optimize server choices based on proximity and load.

Static Router

Ann Arbor, MI

- Built a static router with basic packet forwarding capabilities to route real packets to HTTP servers.
- Implemented layer 2 and layer 3 protocols, including ARP, ICMP, and Ethernet, for routing and handling network traffic.

Embedded Device for Keystroke Timing and Acoustic Attack Protection

Ann Arbor, MI

- Designed the device to intercept keystrokes and introduce random delays before sending to the PC.
- Implemented keystroke sound playback to counter acoustic attacks using recorded sounds.

- Based the system on the STM32F405 microcontroller with Embedded Rust for secure and efficient performance.
- Delivered a compact, user-friendly design with production costs around \$24.50 per unit.
- Utilized SD card for storing custom keystroke sounds and USB peripherals for communication with keyboard and host PC.

PROFESSIONAL SERVICE

- VP 160 Honors Physics SJTU, 2022 Summer

SKILLS

Computer: C++, C, Python, Rust, Java, Pytorch, CUDA, System Verilog, Embedded C/Rust, Linux, MATLAB, Git, LaTeX, LLVM, JAX

HONORS

Dean List, <i>Umich</i>	2023
University Honor, <i>Umich</i>	2022
Tang Junyuan Scholarship, <i>SJTU</i>	2022
SJTU Undergraduate Excellent Scholarship Class B, <i>SJTU</i>	2022